

Fire Detection Using python IDE and open-cv modules

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ABSTRACT

The most common hazard among the different hazards is fire accident (or) fire hazard. So early detection and prevention of the fire accident would cause less casualty and first-aid treatment can be provided on time. This can be implemented using different algorithms and techniques i.e., one among the technique is by using aerial image-based computer vision algorithms like convolution neural networks and image processing methods, here in this case it uses a separable CNNs which is a variant of CNN which converts the raw image into max pool by applying separable conv 2D and Batch normalization to obtain the output layer in image processing unit will detect whether the obtained image has fire or not based upon that it outputs the result, but this is more complex over the operations it performs, to overcome this a proposed method which overcomes the drawbacks of the existing method would be implemented using computer vision technique using algorithms of python IDE which directly deals with the vision of the situation and alerts on time to overcome the situation on time, this method can be implemented at the places where the population density is more and operable on the road side using cameras. And it is also operable using manually or using hardware setup. So, with this methodology we can easily access the scenario and react on time.

KEYWORDS : Convolution Neural Networks (CNN), Computer Vision (Open-cv), Python IDE, fire detection, HSV, masking operation.

1.1 INTRODUCTION:

With the increases of Earth's temperature, the climatic changes be occurring due which it impacts on the environment system and its imbalances the ecology. to overcome this a purposed method which overcome the drawbacks of the existing method would be implemented using computer vision technique using algorithms of python IDE which directly deals with the vision of the situation and alert on time to overcome situation on time ,this method can be implemented at the places were the population density is more and operatable on the road side using cameras. And it also operatable using manually or using hardware setup.so, with this methodology we can easily access the scenario and react on time.

Here with the help of camera module on the road side we can access the scenario and operate .due to which the fire accident can be avoided and immediate action can be taken to prevent the loss of lives of humans and animals .This also be implemented on the forest side in order to avoid the fire accident by the lightning by having a forest observation team to monitor and to immediate action .

There are various approaches be them to implement this working among them few are by using convolution neural networks but the major drawback with this methodology is that complex process to extract the data and again we have to apply the image process techniques to further more .And other method is by deep learning technique and it is also a complex due to the algorithms .

But in this paper, I approached the open-cv a computer visualization technique method with the help of modules of computer vision ,with this we can visualize the real world scenario and detect the fire in it and processing the data with open-cv modules ,and based upon that on time action can be performed .

1.2 LITERATURE SURVEY

T. J. Brown, B. L. Hall, and A. L. Westerling [1] “The impact of twenty first century climate change on wildland fire danger in the western united states: an applications perspective,” *Climatic Change*, vol. 62, no. 1, pp. 365–388, 2004.

States that, High-temporal resolution meteorological output from the Parallel Climate Model (PCM) is used to assess changes in wildland fire danger across the western United States due to climatic changes projected in the 21st century. A business-as-usual scenario incorporating changing greenhouse gas and aerosol concentrations until the year 2089 is compared to a 1975–1996 base period. Changes in relative humidity, especially drying over much of the West, are projected to increase the number of days of high fire danger (based on the energy release component (ERC) index) at least through the year 2089 in comparison to the base period.

Y. Zhang and Z. Xia [2] “Research on the image segmentation based on improved threshold extractions,” in 2018 IEEE 3rd International Conference on Cloud Computing and Internet of Things (CCIoT). IEEE, 2018, pp. 386–389.

States that, As the name suggests, segmentation aims to segregate or segment similar objects in a raw image based on uniformities in object texture, color, and brightness. The scientist in this paper introduces three typical image segmentation methods based on threshold (iterative method, global threshold segmentation, Otsu method, global threshold segmentation, local threshold segmentation), and comparison of results by using computer simulation show the difference of these three methods.

L. N. Smith [3] “A disciplined approach to neural network hyperparameters: Part 1—learning rate, batch size, momentum, and weight decay,” arXiv preprint arXiv:1803.09820, 2018.

States that, how to examine the training validation/test loss function for subtle clues of underfitting and overfitting and suggests guidelines for moving toward the optimal balance point. Then it discusses how to increase/decrease the learning rate/momentum to speed up training. Our experiments show that it is crucial to balance every manner of regularization for each dataset and architecture.

C. Shorten and T. M. Khoshgoftaar [4] “A survey on image data augmentation for deep learning,” *Journal of Big Data*, vol. 6, no. 1, pp. 1–48, 2019

States that, The majority of recently demonstrated Deep-Learning Side-Channel Attacks (DLSCAs) use neural networks trained on a segment of traces containing operations only related to the target subkey. However, when the number of training traces is restricted such as in the ASCAD database, deep-learning models always suffer from overfitting since the insufficient training data.

E. D. Cubuk, B. Zoph, D. Mane, V. Vasudevan, and Q. V. Le [5] “Auto augment: Learning augmentation strategies from data,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 113–123.

States that, we describe a simple procedure called Auto Augment to automatically search for improved data augmentation policies. In our implementation, we have designed a search space where a policy consists of many sub-policies, one of which is randomly chosen for each image in each mini-batch.

2.1 EXISTING METHOD

The CNN algorithm attempts to replicate an eye's visual cortex, which breaks down images into small receptive fields reactive to specific locations in the visual field. CNN requires less computation time than deep neural networks due to its unique spatial independent algorithm that does not require using all available pixels from an input image in the initial layers. Mathematically, a standard convolution operation can be expressed as in

$$y_i = b_i + \sum_{j=1}^{N_i} w_i \cdot k_j^i; j = 1, 2, \dots, N_i \quad \dots \dots \dots (1)$$

To limit overfitting of input data, CNN uses a subsampling layer, called the pooling layer, which aggregates or extracts the maximum value from its input. A CNN architecture begins with an input layer and ends with an output layer. Separable convolution neural networks are a variant of CNN. Unlike a regular CNN architecture, a separable CNN architecture models the spatial features (like shapes) and cross-channel features (like detailed patterns) separately for each available colour channel. Spatial feature extraction in a separable CNN is performed using depth wise convolutional operation and can be mathematically expressed as in, while the extraction of cross-channel features is performed using pointwise convolution operation as shown in

$$y_i = b_i + W_j \cdot k_i; i, j = 1, 2, \dots, N_i \quad \dots \dots \dots (2)$$

Since image features are analysed using two different procedures simultaneously, as mentioned, separable CNN is computationally faster and show better classification properties than the existing CNN architectures.

2.1.1 BLOCK DIAGRAMS:

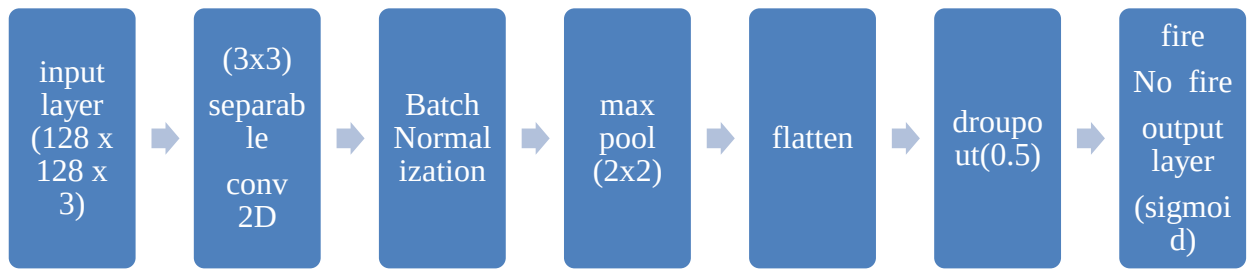


Figure 1 .Architecture of the proposed simple separable convolution neural network

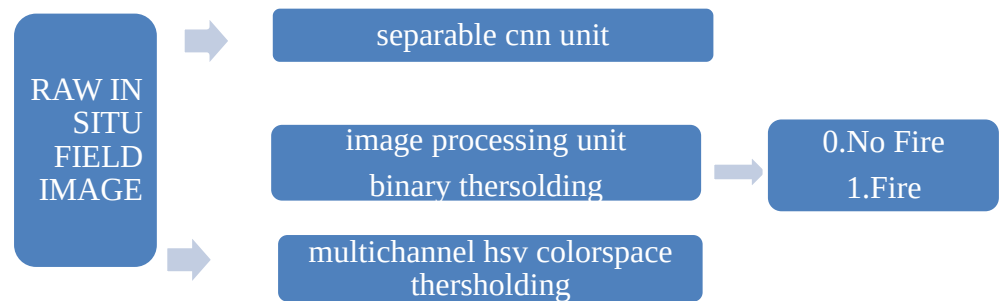


Figure 2 . Proposed combined architecture using separable CNN and digital image processing

2.2 PROBLEM IDENTIFICATION:

- CNN requires less computation time than deep neural networks due to its unique spatial independent algorithm that does not require using all available pixels from an input image in the initial layers
- A simple CNN model with two separable two-dimensional convolution layers showed a recommendable performance when compared to complex models with many convolution layers.
- A model with separable convolutional layers showed greater accuracy with less computational time compared to a regular two dimensional convolution network.
- The finalization of the CNN model was based on performance metrics: AUC, accuracy, precision, recall or sensitivity, and specificity against the test set.
- Further, performance in terms of validation accuracy and computational time against models with one, two, three, four, and six convolution layers were assessed.
- The addition of the max-pooling layer followed by each of the two convolution layers showed better results than the average pooling.

2.3 PROPOSED METHOD :

COMPUTER VISUALIZATION USING OPEN-CV MODULES

Computer visualization main aim is to reduce the operation time complexity and complex algorithms to be reduced by using the open-cv modules which has different functioning libraries used to build an algorithm which can detect the fire with the given parameters that includes masking of the fire image which is processed by the given parameter in the algorithm and the output has been displayed using HSV model and it alerts using siren , displays the 'fire detected' sign .

2.3.1 PROPOSED BLOCK DIAGRAM:

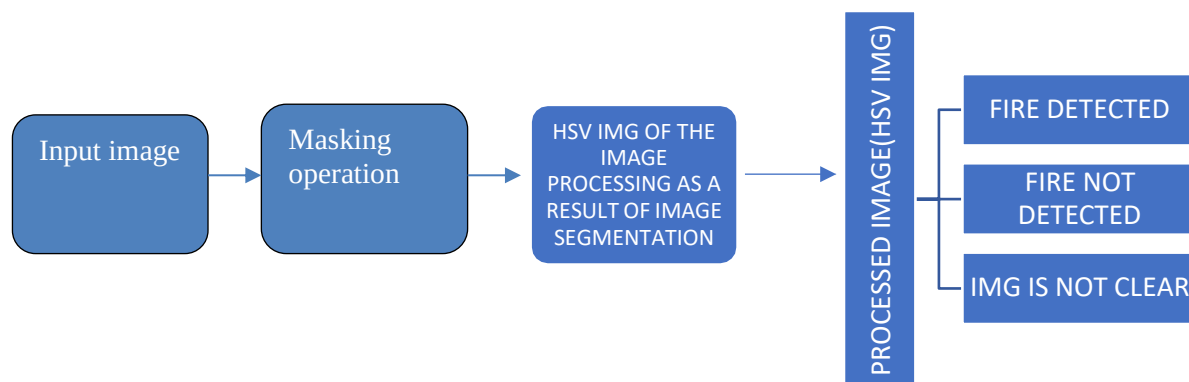


Figure 3 . Proposed block diagram

2.3.2 ALGORITHM:

1. We given input image , video or real-world scenario
2. Then the image be extracted using masking operation and applied HSV model to see the firein the image.
3. Based on the HSV model fire be detected and alerted .

3.1 SOFTWARE USED:

PYTHON (PYCHARM SOFTWARE IDE):

Python is one of the easiest programming languages which can be interpreted easily and understand easily.

NUMPY:

Numpy is one of module of the open-cv package used for the detection algorithms in the real world projects .

OPEN-CV:

Open-cv is computer visizilation module which consists of a huge collection of libraries for image segmentation and image processing techniques

3.1.1 PRACTICAL SETUP:

3.1.2 ALGORITHM:

```
import cv2
import numpy as np
import playsound as ps

fire_reported = 0
alarm_status = False

def play_audio():

ps.playsound("C:\\Users\\Ranjiv\\PycharmProjects\\demo1\\alarm
.mp3")

video = cv2.VideoCapture("video.mov.mp4")
while True:
    ret, frame = video.read()
    frame = cv2.resize(frame, (1000,500))
    blur = cv2.GaussianBlur(frame, (15,15),0)
    hsv = cv2.cvtColor(blur,cv2.COLOR_BGR2HSV)

    lower = [18,50,50]
    upper = [35,255,255]

    lower = np.array(lower,dtype="uint8")
    upper = np.array(upper,dtype="uint8")

    mask = cv2.inRange(hsv,lower,upper)

    output = cv2.bitwise_and(frame,hsv,mask=mask)
    no_of_total = cv2.countNonZero(mask)
    if int(no_of_total)>15000:
        print('fire detected')
        if ret == False:
            break
        cv2.imshow("output", output)
        if cv2.waitKey(10000) & 0xff == ord("q"):
            break
        play_audio()
        break;
    '''if int(no_of_total) > 15000:
        fire_reported = fire_reported + 1
        if fire_reported >= 1:
            if alarm_status == False:
                play_audio()
                alarm_status = True'''
    if ret == False:
        break
    cv2.imshow("output",output)
    if cv2.waitKey(1) & 0xff == ord("q"):
        break
```

```
cv2.destroyAllWindows()  
video.release()
```

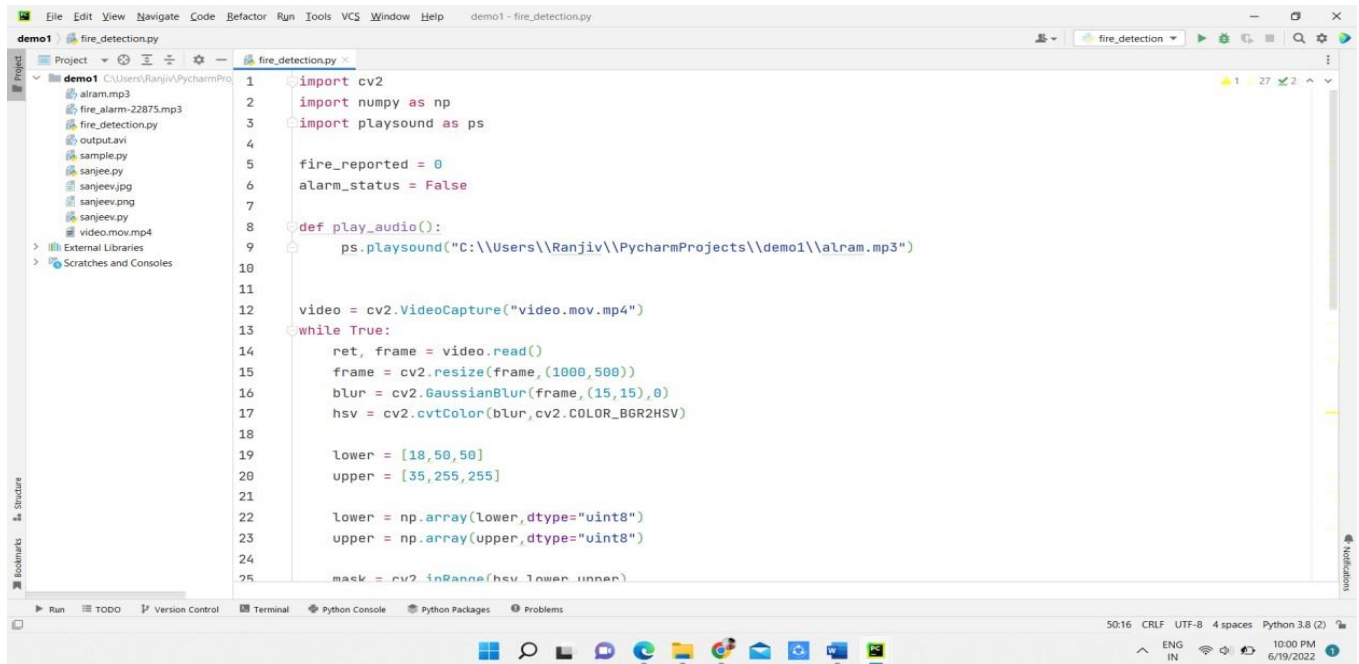


Figure 4 : the practical setup using python – PyCharm editor

4.1 RESULT AND DISCUSSION:

4.1.1 CASE 1:

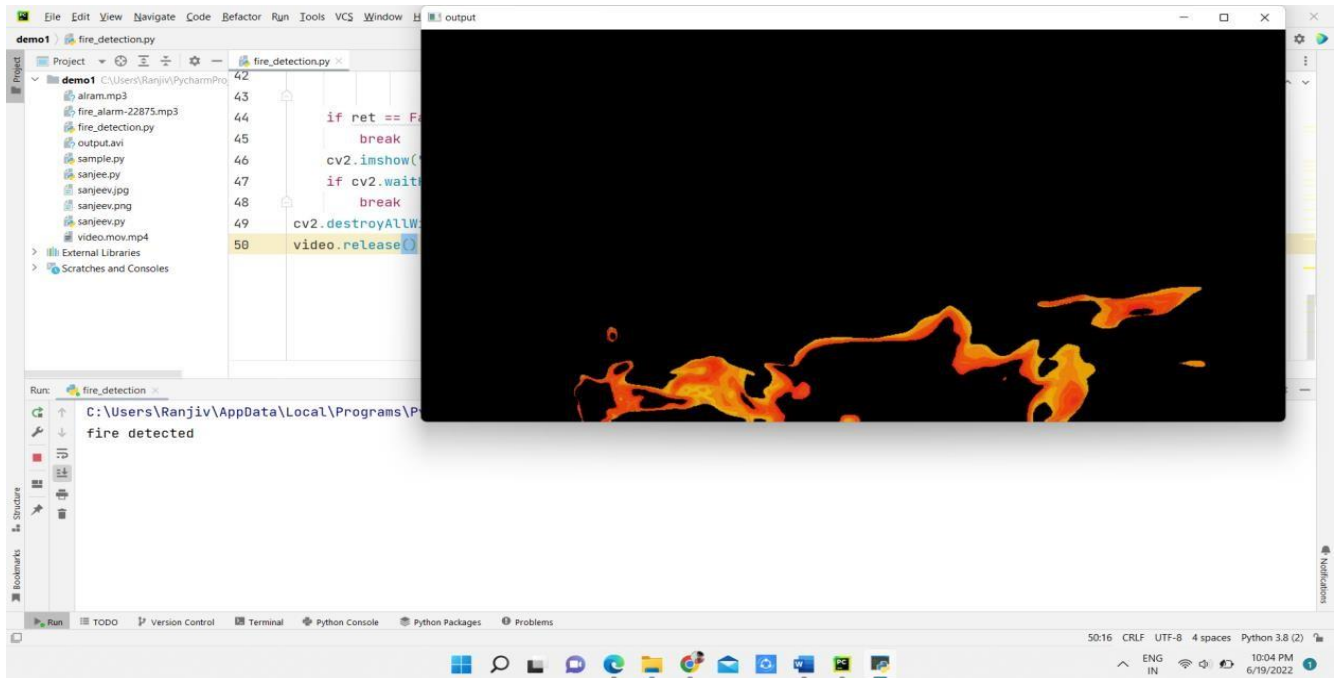


Figure 5 : When a video be used as input it extract an image and process the data ,shows the result. And theoutput is of hsv image.

4.1.2 CASE 2:

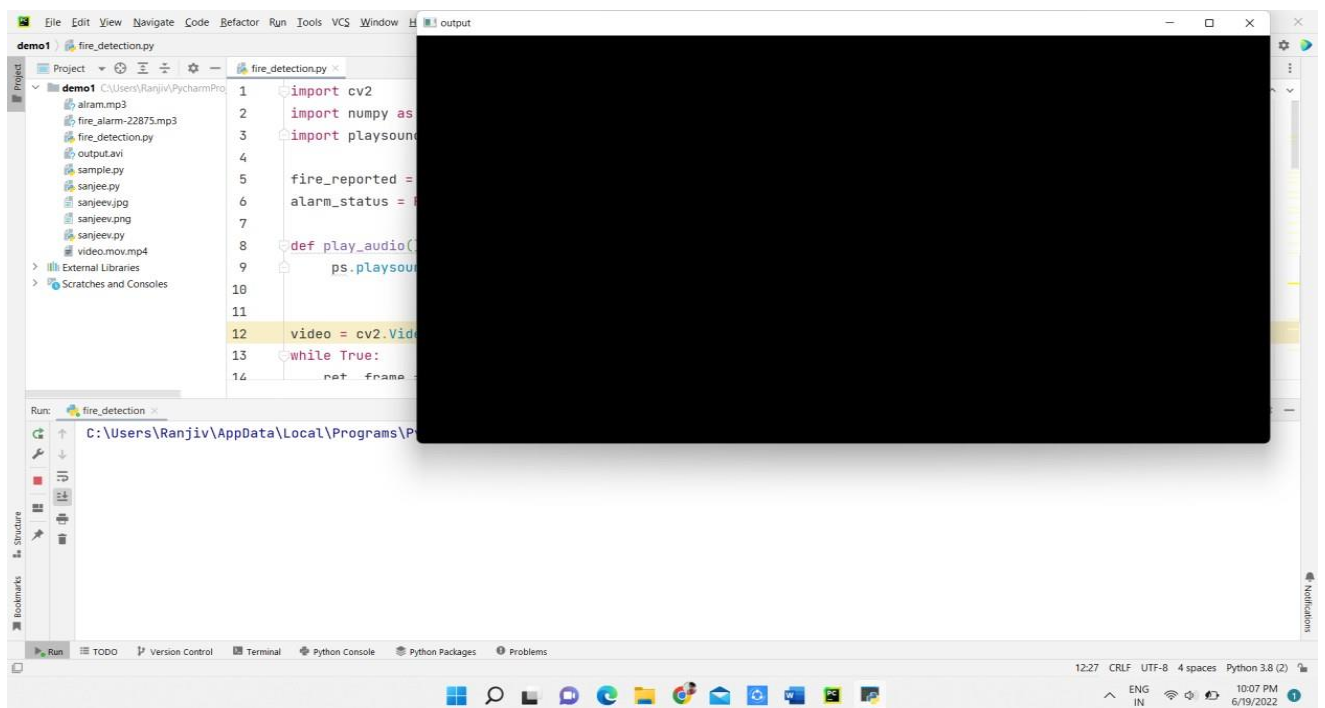


Figure 6: Hsv output of my pc camera as an input given to the algorithm

4.1.3 CASE 3:

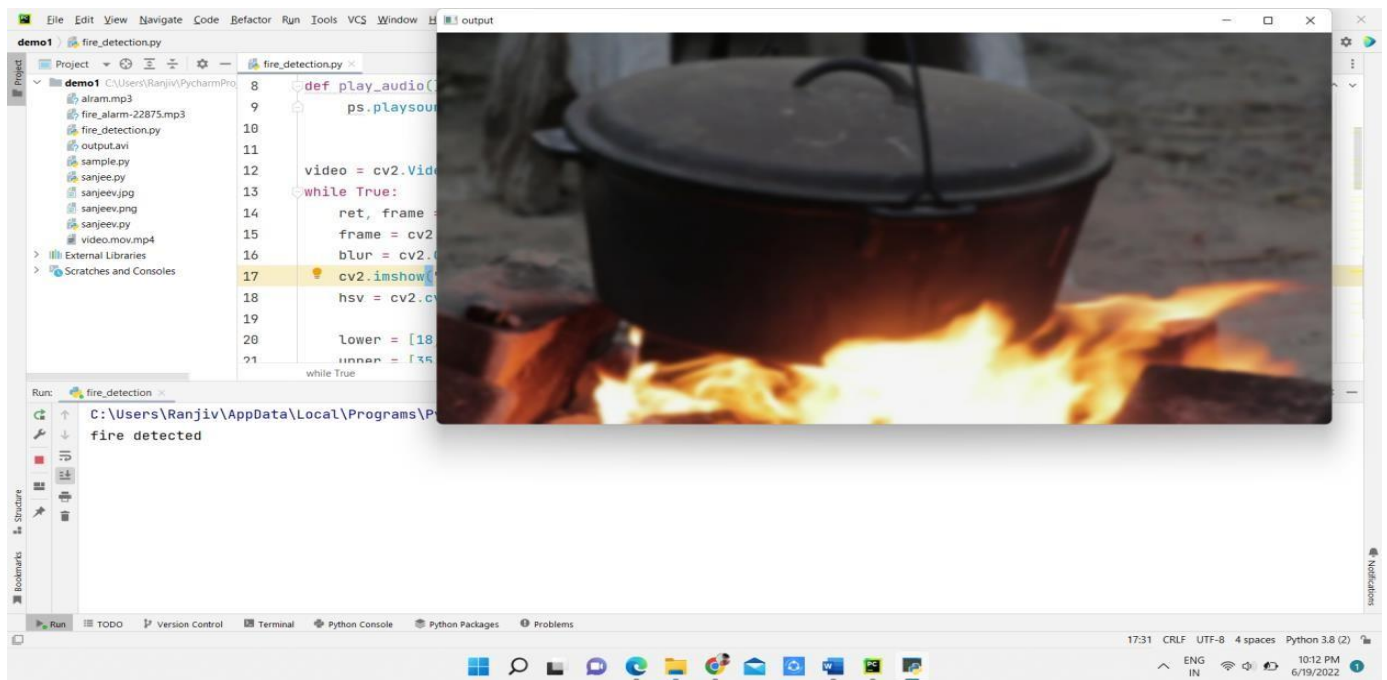


Figure 7: Blur image of the video as an input

4.1.4 CASE 4:

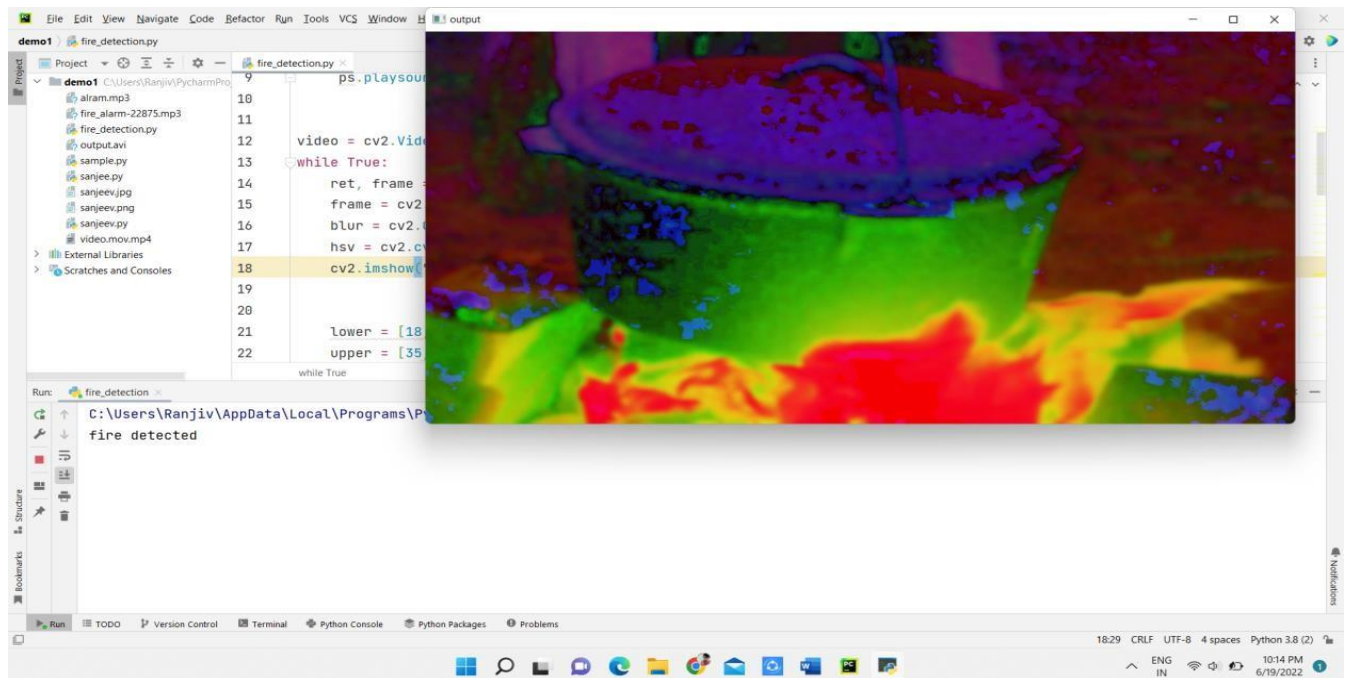


Figure 8: Output result as a hsv of the video image

5.CONCLUSION:

Hence after performing the fire detection using the modules of open-cv in python IDE using PyCharm software .It is 70 – 80% accurate to find the fire in the video ,images and real-world cases and given an output as a result of “fire detected” with fire alarm sounds.

5.1 FUTURE SCOPE :

With this project can control the fire accidents by reacting on time as a first responders to the action

- It can be implemented on the houses , building, industries etc.
- It can be implemented on the forest areas to protect the animals as well as environment from the fire accidents.

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